

Möbius Control for Binary Oppositions: A Field-Based Framework

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August 15, 2025

Abstract

Binary oppositions are formalized as a triad (A, B, T) with differential tension enabling generation, stabilization, and safe inversion. A contour-preserving diffusion with an event-triggered Möbius flip redistributes saturated energy toward void anchors and snaps to a mid-stability point. The resulting two-layer controller co-manages signals (purity, symmetry, emergence) and yields reproducible artifacts for convergence.

Keywords: Möbius topology, binary oppositions, field-based control, symmetry, emergence

1 Introduction

Opposition structures (reason vs. affect, push vs. pull, $0-1-\infty$) are powerful but brittle: once a pole dominates, the system either stalls or oscillates. Standard pipelines that assume separable domains struggle to *invert* a stance safely while keeping coherence. We propose a Möbius-based formulation in which the poles A and B live on a single continuous surface and are connected by a boundary phase. In this view, information is not just a label on states but a field $\mathcal{I}$ that modulates tension between them.

Contributions (1) A compact formalization $\langle A, B, \mathcal{I} \rangle$ with measurable signals: symmetry S , emergence Q , and purity P . (2) A two-layer mechanism: contour-preserving diffusion and an event-triggered Möbius flip with redistribution to void anchors. (3) Operational guidance with figures and minimal artifacts for reproducibility.

2 Theoretical Background

Opposition as structure. Classical binary models separate poles sharply; a Möbius surface renders them contiguous: traversing the band reverses orientation without leaving the surface. **Information as field.** $\mathcal{I}(x, y, t)$ encodes local tension: large magnitude indicates saturation at a pole; small values indicate balance near the midpoint. **Auditable signals.** We monitor symmetry S , emergence Q , and purity P to trigger and evaluate inversions.

3 Methodology

(L1) Contour-preserving diffusion

Between events we diffuse $\mathcal{I}$ while preserving edges, spreading energy away from hotspots without erasing structure.

(L2) Event-triggered Möbius flip

When saturation exceeds a threshold and purity P is low, we flip orientation and redistribute mass toward void anchors. After recovery (drop in $\max|\mathcal{I}|$ and rebound in P), we return to diffusion.

4 Results

Figures ??–?? summarize a synthetic run.

Before→after→recovery (Fig. ??). At t_0 the field saturates near a pole; at t_1 the flip carves a low-tension channel and drains the hotspot; by t_2 diffusion rebalances the domain.

Timeline (Fig. ??). Immediately after the flip, $\max|\mathcal{I}|$ drops; symmetry S and purity P rebound; emergence Q rises—the desired “safe novelty” profile.

Schematic (Fig. ??). The thought-galaxy is enclosed by a Möbius shell. Stars are void anchors; the double-headed arrow denotes self-inversion around the midpoint.

5 Discussion

The controller exposes auditable signals and an explicit event boundary, allowing policy rules (e.g., flip only when $P < \epsilon$ and cooldown satisfied). Because the surface is single-sided, stance changes are reversible without introducing a second disconnected state space. Limitations include domain-specific calibration of thresholds and metrics.

6 Conclusion

We presented a field-based, Möbius-topological approach to binary opposition that enables safe, auditable inversions while retaining coherence. Future work: (i) learn anchor placement from data, (ii) couple $\mathcal{I}$ to task objectives, (iii) multi-agent extensions.

Data & code availability [mobius-opposition-kit.zip](#).

A short README describing the folder layout and how to run the demo is here: [README_en.txt](#).

Direct URLs

<https://drive.google.com/file/d/1XUganKIOuru552QjbbAoGaHcVw3gwMjb/view?usp=drivesdk>

<https://drive.google.com/file/d/1NKpBHVssaseE84RJuoF9bnKFEVL4ZVLA/view?usp=drivesdk>